

Tristan Lazard

37 Boulevard Mortier
75020, Paris
@trislaz

+336.89.95.84.82
trislaz.tl@gmail.com
trislaz.github.io

EDUCATION	PSL University , Paris, 75006 <i>ITI - MS Innovation</i> , Entrepreneurship and innovation.	2018-2019.
	Sorbonne University , Paris <i>Master 2 : Applied mathematics</i> Thesis: "A model of random mutualistic network."	2016-2017
	Ecole Normale Supérieure , rue d'Ulm, Paris <i>Master 1 : Evolutionary biology and ecology</i> <i>Bachelor in life-sciences</i>	2015-2016 2013-2014
	Lycée Saint-Louis , Paris <i>Classe préparatoire BCPST</i>	2011-2013
	AgroCampus-Ouest , Rennes <i>Bachelor in Agronomy, spare year</i>	2014-2015
SKILLS	Languages: French (mother tongue), fluent English and debutant in Spanish. Programming languages: Python, Bash, $\text{\LaTeX}$, nextflow. Computing skills: docker, slurm, cloud-computing, GPU-computing, ML+image processing usual libs (pytorch, openCV, PIL, sklearn...).	
PROFESSIONAL EXPERIENCE	PhD candidate December 2019 - December 2023 My PhD project focuses on processing Whole Slide Images, which are digitized microscopic images of diseased tissue commonly used by clinicians for diagnosis. These images contain a vast amount of information, leading to large file sizes (10-15 GB uncompressed). By developing machine learning algorithms to predict patient-level labels from these images, we can assist clinicians in their practice and improve our understanding of the disease through the interpretation of the algorithm's decisions. Main Achievements: • 5 publications, 3 first authors, in biological journals and machine learning venues. • 1 patent. • Prize-winning solution (5000\$) at the VisioMel challenge. Link to the solution	CBIO - Mines-Paristech Paris, France
	Engineering internship Jan - July 2019 <i>Automatic counting of Melanocytes</i> : In partnership with L'Oréal, I created an algorithm for automatically counting melanocytes in microscopic images. This solution has been integrated into the processing pipeline and is now utilized in regular research.	CMM, Mines-Paristech Fontainebleau, France
	Research internship Jan - August 2018 <i>Large scale phylogenetic tree inference</i> : Develop an algorithm for calculating phylogenetic trees from huge genetic sequence data (-omics data).	IBPS, Sorbonne university Paris, France

PUBLICATIONS **Lazard, T.**, Bataillon, G., Naylor, P., Popova, T., Bidard, G.-C., Stoppa-Lyonnet, D., Stern, M.- H., Decencie‘ re, E., Walter, T., and Vincent-Sal- omon, A. (2022). Deep Learning identifies new morphological patterns of Homologous Recombination Deficiency in luminal breast cancers from whole slide images. **Cell Reports Medicine**

Lazard, T., Lerousseau, M., Decenci re, E., Walter, T. (2023). Giga-SSL: Self-Supervised Learning for Gigapixel Images. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**) **Workshops**.

Lazard, T., Blusseau S., Velasco-Forero S, Decenci re E., Flouret V., Cohen C., Baldewick T. (2022). Applying Deep Learning to Melanocyte Counting on Fluorescent TRP1 Labelled Images of In Vitro Skin Model. **Image Analysis Stereology**.

Lubrano M., **Lazard T.**, Balezo G., Bellahsen-Harrar Y., Badoual C., Berlemont S., Walter T. (2022). **Computer Vision–ECCV 2022 Workshops**: Tel Aviv, Israel, October 23–27, 2022, Proceedings.

Naylor P, **Lazard T**, Bataillon G, La  M, Vincent-Salomon A, Hamy AS, et al. Prediction of Treatment Response in Triple Negative Breast Cancer From Whole Slide Images. **Front Signal Process**.

Lazard T., Decenci re E., Walter T. Democratizing computational pathology: optimized WSI representations for the TCGA. **Preprint**.

INTERESTS Climbing, saxophone and harmonica, poetry and literature.